7.2.2. Expectation and variance

There are a number of questions we can ask a probability distribution. We can ask what the single "most likely" outcome is, which we commonly think of as the *mean* or *average* of the distribution. You're probably familiar with the calculation of the mean as taking the sum of all the results and dividing by the number of results. For example, the mean of the 5-day temperature forecast of [18, 21, 17, 17, 21]°C is [18 + 21 + 17 + 17 + 21]/5 = 94/5 = 18.8°C. This is the average predicted temperature over a sample of 5 days in Chicago, Illinois, USA.

Consider if instead we asked five people to give us their prediction for tomorrow's temperature in Chicago and they happened to give us the same numbers, [18, 21, 17, 17, 21]°C. If we wanted the average temperature for tomorrow, we would follow the same procedure, adding the numbers up and dividing by the number of samples (five) to get the average predicted temperature for tomorrow. But what if person 1 was a meteorologist, and we had a lot more confidence in their prediction compared to the other four people that we randomly polled on the street? We would probably want to weight the meteorologist's prediction higher than the others. Let's say we think that their prediction is 60% likely to be true, and the other four are merely 10% likely to be true (notice 0.6 + 4 * 0.10 = 1.0), this is a *weighted average*; it's computed by multiplying each sample by its weight. In this case, that works out as follows: [(0.6 * 18) + 0.1 * (21 + 17 + 17 + 21)] = 18.4°C.

Each temperature is a possible outcome for tomorrow, but not all outcomes are equally likely in this case, so we multiply each possible outcome by its probability (weight) and then sum. If all the weights are equal and sum to 1, we get an ordinary average calculation, but many times it is not. When the weights are not all the same, we get a weighted average called the *expectation value* of a distribution.

The expected value of a probability distribution is its "center of mass," the value that is most likely on average. Given a probability distribution, $P(x)$, where x is the sample space, the expected value for discrete distributions is calculated as follows.